

A Unified Control Plane for Machine-Learning Security Services

An Authenticated Gateway for Independently Released ML Security Products

Evidence status: Refreshed against current code snapshot 01e58ba257764e47e230a3eecd5a821bb85e7985 and successful CI/CD run 36783840279 on 2026-09-30.

Contribution

A real authenticated FastAPI control plane with four production HTTP service slots, a local contract-test topology built from stubs, and a production Compose contract that requires externally released product images instead of pretending the stubs are product implementations.

Method

1. Authenticate external callers at the gateway.
2. Strip caller-controlled sensitive headers and inject per-service credentials.
3. Route four synchronous service prefixes over an internal-only network.
4. Use local health-contract stubs to test topology without misrepresenting product logic.
5. Require explicit external product images in production Compose.
6. Validate unit, security, integration, image-build, and deployment-contract behavior in CI.

Current Verified Evidence

Current-main Python 3.12 unit-test job reports:

- 67 tests passed.
- 56.89% statement coverage.
- Separate local contract-stub health jobs for MCP, LLM red-team, dataset-poison, and model-privacy completed successfully.
- Integration topology tests completed successfully.
- Security scan, dependency audit, production Compose validation, and gateway image build/push job completed successfully.

Reproducibility

Expected core Python evidence: 67 passed, 56.89% coverage.

Honest Boundaries

- Product business logic is owned by external repositories/images.
- No production deployment, uptime, or load SLO is established.
- Multi-replica shared rate limiting/state distribution requires deployment-specific infrastructure.
- Batch tools such as HF provenance scanning and adversarial robustness are intentionally not forced behind synchronous HTTP routes.